

Master's Final Semester Project Proposal

StudentOS: A Data-Driven Student Decision Intelligence Platform

1. Introduction

University students are required to make numerous academic and career-related decisions throughout their degree, including selecting courses, planning their semesters, identifying suitable career paths, developing relevant skills, choosing projects and preparing for employment. However, the information required to make these decisions is often scattered across university course catalogs, degree requirements, student reviews, job postings, career websites and other disconnected sources.

Most existing student platforms focus primarily on administrative functions such as course registration, grades, attendance or access to learning materials. Career platforms on the other hand, generally focus on job searching and do not consider a student's academic history and completed coursework in sufficient depth. As a result, students often have to manually connect their academic experiences with their future career goals.

This project **StudentOS**, a data-driven web application designed to act as a student decision intelligence platform. The system will integrate academic, course, skill and career-market data into a unified data platform and use this information to provide students with personalized academic and career insights.

The primary objective is not to replace university systems but to create an intelligent layer on top of existing information that helps students understand what they have already achieved, what they are missing and what they should do next.

2. Problem Statement

Students currently face several challenges when making academic and career decisions.

First, course information is generally presented independently. A course catalog may describe the content and prerequisites of a course, but it does not necessarily explain how that course contributes to a student's long-term career objectives.

Second, students have limited visibility into the relationship between courses, skills, projects and employment opportunities. For example, a student interested in becoming a Data Engineer may know that they should learn SQL or Python, but may not know which courses, projects or technologies would best prepare them for that career.

Third, job descriptions contain valuable information about the skills demanded by employers, but this information is rarely connected directly to university curriculum. Consequently, students may graduate with a degree but discover that their skill profile does not fully align with the current job market.

This project aims to address these issues by creating a unified data ecosystem connecting:

Students -> Courses -> Skills -> Projects -> Technologies -> Careers -> Job Opportunities

The system will transform these disconnected sources into actionable recommendations and visual insights.

3. Proposed Solution

StudentOS will be developed as a web-based platform where students create or import an academic profile containing information such as their degree, completed courses, grades, technical skills, projects, interests and career objectives.

The system will then analyze this information against a structured knowledge base containing course information, prerequisite relationships, skills associated with courses and current job-market requirements.

For example, a student pursuing a career in Data Engineering may have completed courses in databases, cloud computing and machine learning. StudentOS could identify that the student already possesses strong SQL and cloud-related knowledge but has limited exposure to technologies such as Apache Kafka, Apache Airflow or Terraform.

The platform could subsequently recommend:

- Courses that address the student's skill gaps.
- Projects that would strengthen the student's portfolio.
- Technologies that should be learned.
- Career paths that match the student's existing profile.
- Job categories for which the student is currently qualified.
- Skills that would increase their employability.

The goal is to provide evidence-based recommendations rather than generic career advice.

4. Key Features

4.1 Student Academic Profile

Students will be able to create/upload a profile containing:

- Degree and major
- Completed and planned courses
- Academic performance
- Technical skills
- Projects
- Certifications
- Career interests
- Target career

The system will convert this information into a structured student profile.

4.2 Course Intelligence

The platform will maintain relationships between courses, prerequisites, topics and skills.

For example:

Data Engineering -> Distributed Systems -> Cloud Computing -> Data Engineer

Students will be able to explore how individual courses contribute to their desired career paths.

4.3 Career and Skill Analysis

It will analyze job-market data to identify the skills commonly requested for different positions.

A student could receive an analysis such as:

Target Career: Data Engineer

- SQL; Strong
- Python; Strong
- Cloud Computing; Strong
- Apache Spark; Moderate
- Redis; Weak
- Airflow; Weak
- Snowflake; Missing

The platform can then recommend learning activities based on these gaps.

4.4 Course Recommendation

The system will recommend courses based on the student's:

- Completed coursework
- Prerequisites
- Career objectives
- Existing skills
- Skill gaps
- Potential workload

This will make course selection more personalized than simply displaying a university course catalog.

4.5 Career Match Score

The system will calculate a student's approximate compatibility with different career paths based on their skills, coursework, projects and job-market requirements.

For example:

Career	Estimated Match
Data Engineer	87%
Cloud Engineer	76%
Backend Engineer	71%
ML Engineer	64%

The score will be accompanied by an explanation of the factors contributing to it rather than being presented as an unexplained prediction.

5. Data Engineering Component

A major objective of the project is to ensure that StudentOS is not simply a conventional web application with a recommendation algorithm. The core of the project will be a **data engineering architecture** capable of collecting, processing, integrating and analyzing heterogeneous datasets.

Potential data sources will include:

- University course catalogs
- Degree requirements

- Course prerequisites
- Public job postings
- Technology/skill taxonomies
- Student-generated course feedback
- Public career information
- Synthetic academic records for testing

The proposed architecture will use a simple batch-oriented data pipeline:

Data Sources -> Python Data Ingestion -> Data Cleaning & Transformation -> PostgreSQL Database -> Recommendation Engine -> Web Application

The project can be implemented primarily using Python, PostgreSQL and a web development framework, with tools such as Pandas and SQL for data processing. Docker may optionally be used to simplify deployment and development.

The data pipeline will perform tasks such as:

- Data collection from APIs, CSV files and publicly available datasets
- Data cleaning and validation
- Schema normalization
- Duplicate detection
- Combining information from different sources
- Transforming raw data into structured datasets
- Updating datasets when new information becomes available
- Creating relationships between courses, skills, projects, and careers
- Generating data required by the recommendation system

6. Knowledge Graph

One of the distinctive components of the project will be a student academic and career knowledge graph.

Instead of storing courses, skills and jobs as isolated records, the system will model relationships between them. For example:

Course -> teaches -> Skill

Skill -> required by -> Job

Course -> prerequisite of -> Course

Project -> demonstrates -> Skill

Student → completed → Course

This allows the platform to answer more complex questions.

For example: 'I want to become a Data Engineer. Which courses should I take to close my current skill gaps?'

The system can traverse the relationships between the student's existing skills, university courses and job requirements to generate a recommendation.

7. Research Component

The project will also investigate whether integrating academic data with real-world employment data can produce more useful student recommendations than traditional course-planning systems.

Potential research questions include:

1. How can heterogeneous academic and employment datasets be integrated into a unified student decision support platform?
2. Can a knowledge graph effectively represent relationships between university courses, technical skills, projects and career opportunities?
3. Can student-specific skill-gap analysis provide more relevant course and career recommendations than generic recommendations?
4. How accurately can the system identify the skills required for a student's selected career based on current job-market data?

8. Expected Outcome

The expected outcome is a functional web application that allows students to understand their academic and professional development through a single platform.

A completed prototype should allow a student to:

1. Build an academic profile.
2. Select a target career.
3. Analyze their existing skills.
4. Identify skill gaps.
5. Explore relevant university courses.
6. Receive personalized recommendations.
7. Explore projects and technologies relevant to their career.
8. Understand how their coursework connects to employment opportunities.
9. Visualize their academic and career progression.

The project will therefore combine Web Development, Data Engineering, Data Analytics, Knowledge Graphs and Recommendation Systems into one integrated system.

9. Significance and Innovation

The primary innovation of my project is its attempt to connect information that is normally separated across different systems.

Rather than asking 'What course should I take?', StudentOS attempts to answer; 'Given what I have studied, what I know, what I want to become and what employers currently require, what should I do next?'. This creates a more comprehensive decision-support system for students.

The project is also designed to be extensible. Although the initial implementation may focus on GVSU's CS department, the underlying data architecture can eventually support multiple institutions and different academic disciplines.

10. Conclusion

StudentOS proposes a data-driven approach to improving academic and career decision-making for university students. By integrating academic information, course structures, student profiles, technical skills and employment-market data, the system will provide students with personalized insights into their academic and professional development.

From a technical perspective, the project provides substantial opportunities to investigate modern Data Engineering concepts including data pipelines, distributed processing, data lakehouses, data integration, knowledge graphs, data quality and recommendation systems.

The final product will be both a practical student-facing web application and a research-oriented Data Engineering project, with the potential to demonstrate how interconnected educational and employment data can be transformed into meaningful, personalized decision support for students.